



C23-A-AA-C-M-MET-MNG-MRAC-102

23005

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

FIRST YEAR (COMMON) EXAMINATION

ENGINEERING MATHEMATICS—I

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :** (1) Answer **all** questions.
(2) Each question carries **three** marks.
(3) Answer should be brief and straight to the point.

1. If $f(x) = x^2$, where x is an integer satisfying $-3 \leq x \leq 3$, then find its range.
2. Resolve $\frac{x}{(x-3)(x+2)}$ into partial fractions.
3. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$, then find $3A + 2B$.
4. If $A = \begin{bmatrix} 2 & -4 \\ -3 & 3 \end{bmatrix}$, then find $A + A^T$.
5. Prove that $\frac{\cos 36^\circ + \sin 36^\circ}{\cos 36^\circ - \sin 36^\circ} = \tan 81^\circ$
6. Prove that $\frac{1 + \cos 2\theta}{\sin 2\theta} = \cot \theta$

7. Find the multiplicative inverse of the complex number $(3 + 4i)$.
8. Find the distance between the parallel lines $2x + 3y + 5 = 0$ and $2x + 3y + 9 = 0$
9. Evaluate $\lim_{x \rightarrow 2} \frac{x^5 - 32}{x - 2}$
10. Find the derivative of $3x^2 + 4x - 5$ w.r.t. x .

PART—B

10×5=50

Instructions : (1) Answer *any five* questions.
 (2) Each question carries **ten** marks.
 (3) Answer should be comprehensive and criteria for valuation is the content but not the length of the answer.

11. Solve the equations $2x - y + 3z = 8$, $-x + 2y + z = 4$ and $3x + y - 4z = 0$ by using Cramer's rule.
12. (a) Prove that $\frac{\sin 12A + \sin 6A}{\cos 12A + \cos 6A} = \tan 9A$
- (b) Prove that $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) = \frac{\pi}{4}$.
13. (a) Solve $\cos^2 \theta - 2\cos \theta + 1 = 0$
- (b) Find the area of ΔABC if $a = 5$, $b = 7$ and $C = 30^\circ$
14. (a) Find the centre and radius of the circle $x^2 + y^2 + 4x - 6y - 20 = 0$

- (b) Find the vertex, equation of axis, focus and equation of latus rectum of the parabola $y^2 = 32x$.
- 15.** (a) Find the derivative of $4 \sin x + \log x + 2 \sec x + 5 \tan x$ w.r.t. x .
- (b) Find the derivative of $x^2 e^x$ with respect to x .
- 16.** (a) If $y = x^x$, then find $\frac{dy}{dx}$.
- (b) Find $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$ for $u(x, y) = x \sin y + y \cos x$.
- 17.** (a) Find the equations of tangent and normal to the curve $y = x^2 - 3x - 1$ at $(1, -3)$.
- (b) If $s(t) = 25t + t^3$ is the displacement of a particle, then find its velocity and acceleration at $t = 2$ sec.
- 18.** (a) Find the maximum or minimum value of the function $f(x) = x^2 - 4x + 4$.
- (b) The radius of a circular plate is measured with an error of 2%, find the approximate percentage error in its area.

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